

EC4060 – COMPUTER AND DATA NETWORK

ASSIGNMENT 1

INDEPENDENT LEARNING AND IMPLEMENTATION ASSIGNMENT

DESHAPRIYA D.M.K.N

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1. Network Design Details

Introduction

This project aims to design and simulate a robust and secure network infrastructure for the Faculty of Engineering at the University of Jaffna. The network will cater to diverse needs, supporting seamless communication among academic departments, administrative offices, staff, and students. By applying industry best practices in **subnetting** and **VLAN configuration**, the network will ensure **logical separation** for security purposes while enhancing **network performance** and future scalability. Using **Cisco Packet Tracer**, the network will be simulated and validated for **connectivity**, **reliability**, and **scalability** under real-world conditions. The final deliverables will include a comprehensive **network design**, a **subnet calculation table**, **configuration scripts**, and a **detailed report** outlining the results of the simulation and performance testing.

Objective

1. **Design and simulate a highly scalable and secure network infrastructure** tailored for a multi-branch institution, ensuring optimal performance and robust security for all users.
2. **Implement effective subnetting and VLAN strategies** to create logical divisions within the network, ensuring efficient data routing, secure communication, and room for future network growth.
3. **Test and validate the network design** through comprehensive simulation in **Cisco Packet Tracer**, focusing on connectivity, functionality, and performance to meet institutional needs.

1.1 Subnetting Calculations

Department	Device Type	Total Devices	Future Growth (30%)	Required Hosts	Subnet Mask	Network Address	Usable IP Range	Broadcast Address
Computer Eng	Students	250	325	512	255.255.254.0 (/23)	192.168.0.0	192.168.0.1 - 192.168.1.254	192.168.1.255
	Staff	50	65	128	255.255.255.128 (/25)	192.168.2.0	192.168.2.1 - 192.168.2.126	192.168.2.127
	Other Devices	27	35	64	255.255.255.192 (/26)	192.168.2.128	192.168.2.129 - 192.168.2.190	192.168.2.191
EE Eng	Students	150	195	256	255.255.255.0 (/24)	192.168.3.0	192.168.3.1 - 192.168.3.254	192.168.3.255
	Staff	50	65	128	255.255.255.128 (/25)	192.168.4.0	192.168.4.1 - 192.168.4.126	192.168.4.127
	Other Devices	17	22	32	255.255.255.224 (/27)	192.168.4.128	192.168.4.129 - 192.168.4.158	192.168.4.159
Civil Eng	Students	75	98	128	255.255.255.128 (/25)	192.168.5.0	192.168.5.1 - 192.168.5.126	192.168.5.127
	Staff	25	33	64	255.255.255.192 (/26)	192.168.5.128	192.168.5.129 - 192.168.5.190	192.168.5.191
	Other Devices	7	10	16	255.255.255.240 (/28)	192.168.5.192	192.168.5.193 - 192.168.5.206	192.168.5.207
Mech Eng	Students	75	98	128	255.255.255.128 (/25)	192.168.6.0	192.168.6.1 - 192.168.6.126	192.168.6.127
	Staff	25	33	64	255.255.255.192 (/26)	192.168.6.128	192.168.6.129 - 192.168.6.190	192.168.6.191
	Other Devices	12	16	32	255.255.255.224 (/27)	192.168.6.192	192.168.6.193 - 192.168.6.222	192.168.6.223
IDS	Students	15	20	32	255.255.255.224 (/27)	192.168.7.0	192.168.7.1 - 192.168.7.30	192.168.7.31
	Staff	25	33	64	255.255.255.192 (/26)	192.168.7.32	192.168.7.33 - 192.168.7.94	192.168.7.95
	Other Devices	7	10	16	255.255.255.240 (/28)	192.168.7.96	192.168.7.97 - 192.168.7.110	192.168.7.111
Administration	Staff	25	33	64	255.255.255.192 (/26)	192.168.8.0	192.168.8.1 - 192.168.8.62	192.168.8.63
	Printers	5	7	16	255.255.255.240 (/28)	192.168.8.64	192.168.8.65 - 192.168.8.78	192.168.8.79
CCTV System	CCTV Cameras	50	65	128	255.255.255.128 (/25)	192.168.9.0	192.168.9.1 - 192.168.9.126	192.168.9.127

1.2 VLAN Plan and Mapping

Department	VLAN ID	Device Type	Subnet	Subnet Mask	Network Address
Computer Eng	10	Students	192.168.0.0/23	255.255.254.0	192.168.0.0
	20	Staff	192.168.2.0/25	255.255.255.128	192.168.2.0
	30	Other Devices	192.168.2.128/26	255.255.255.192	192.168.2.128
EEE	40	Students	192.168.3.0/24	255.255.255.0	192.168.3.0
	50	Staff	192.168.4.0/25	255.255.255.128	192.168.4.0
	60	Other Devices	192.168.4.128/27	255.255.255.224	192.168.4.128
Civil Eng	70	Students	192.168.5.0/25	255.255.255.128	192.168.5.0
	80	Staff	192.168.5.128/26	255.255.255.192	192.168.5.128
	90	Other Devices	192.168.5.192/28	255.255.255.240	192.168.5.192
Mech Eng	100	Students	192.168.6.0/25	255.255.255.128	192.168.6.0
	110	Staff	192.168.6.128/26	255.255.255.192	192.168.6.128
	120	Other Devices	192.168.6.192/27	255.255.255.224	192.168.6.192
IDS	130	Students	192.168.7.0/27	255.255.255.224	192.168.7.0
	140	Staff	192.168.7.32/26	255.255.255.192	192.168.7.32
	150	Other Devices	192.168.7.96/28	255.255.255.240	192.168.7.96
Admin	160	Staff	192.168.8.0/26	255.255.255.192	192.168.8.0
	170	Printers	192.168.8.64/28	255.255.255.240	192.168.8.64
CCTV	180	CCTV Cameras	192.168.9.0/25	255.255.255.128	192.168.9.0

TABLE02: VLAN plan and Mapping

1.3 Topology diagram

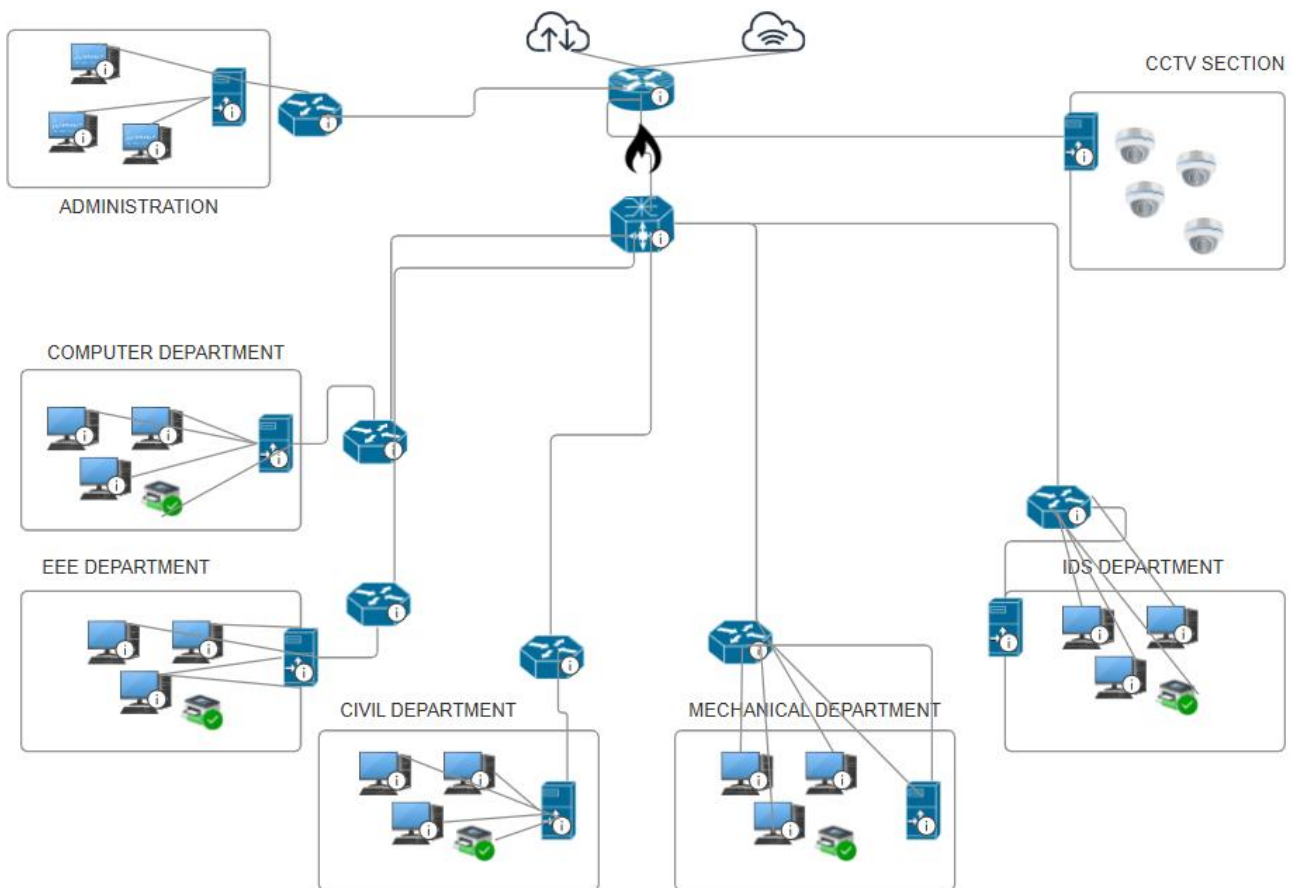
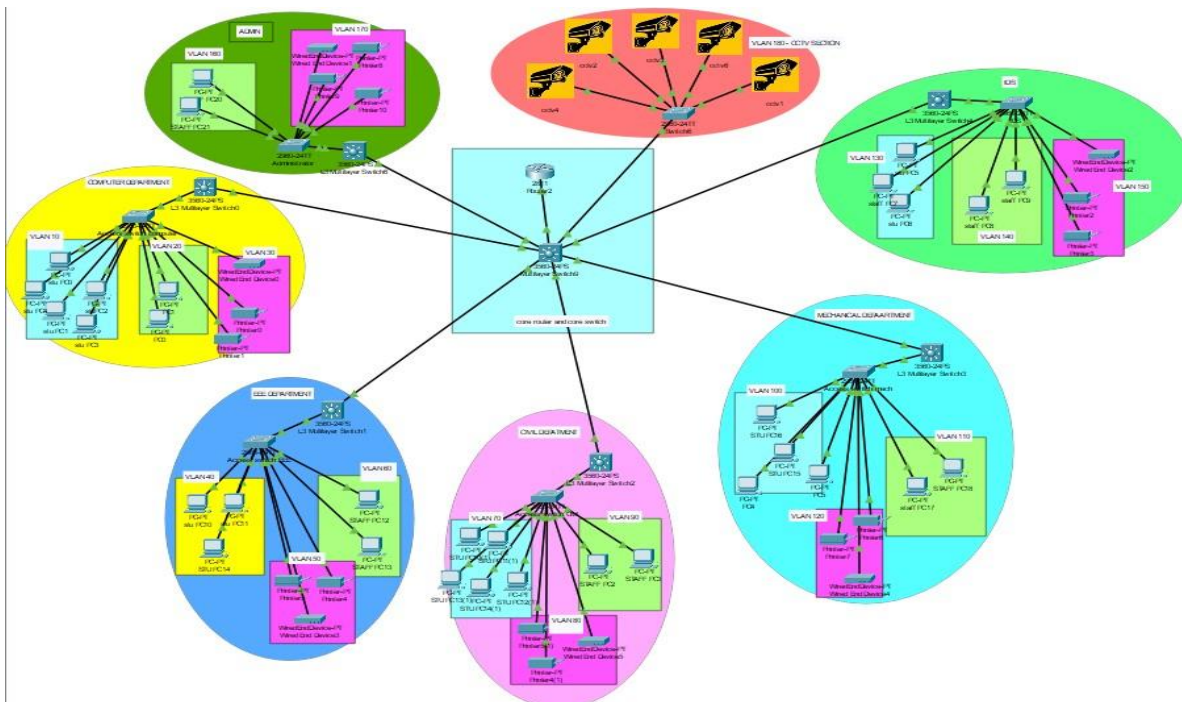


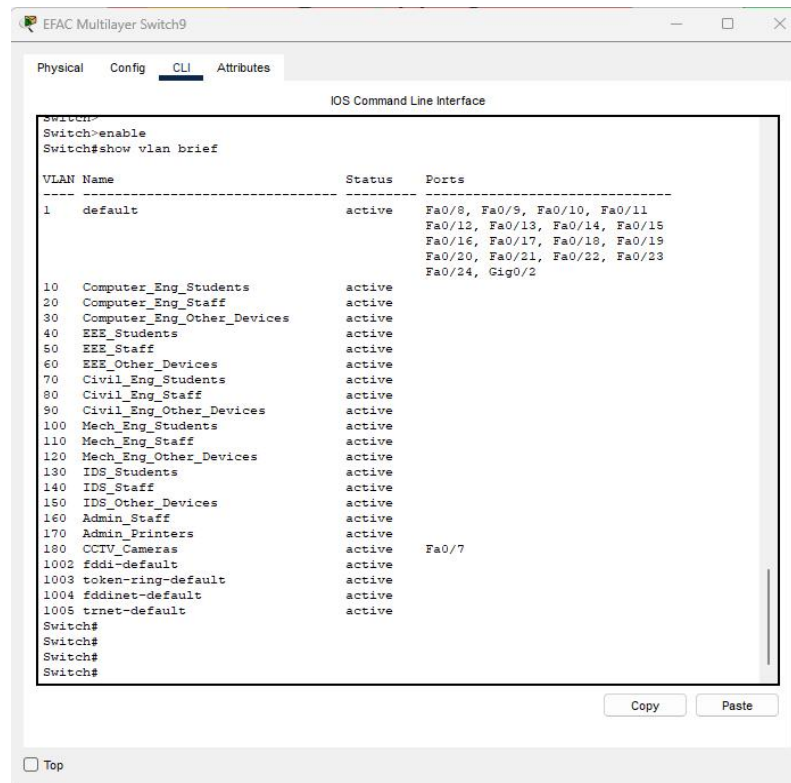
FIGURE01: Topology Diagram



2. Simulation Results:

2.1 VLAN PARTITION

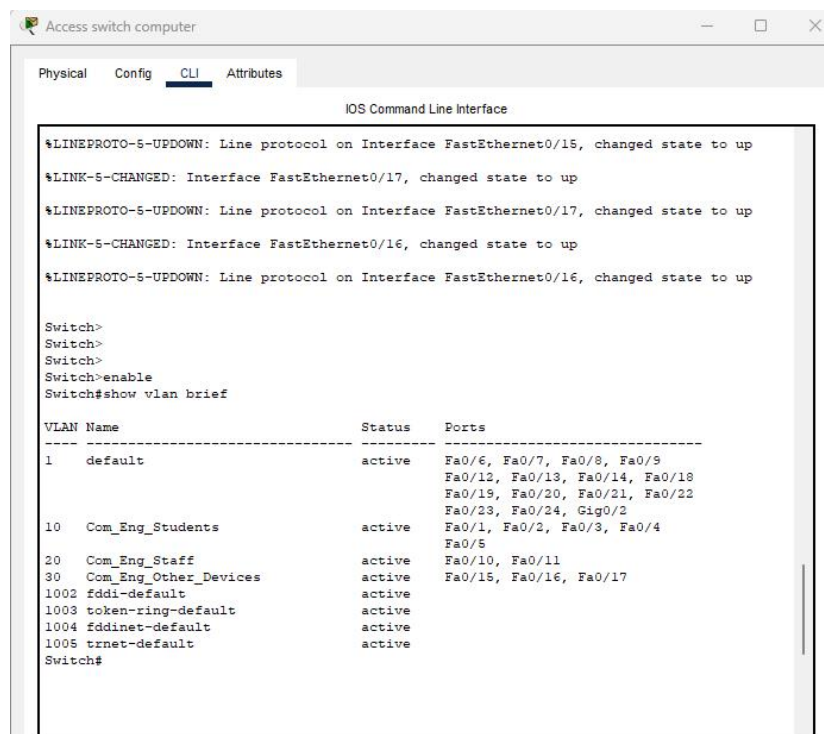
1. Main Multi-layer L3 switch



The screenshot shows the CLI of a switch named 'EFAC Multilayer Switch9'. The user has entered 'enable' and then 'show vlan brief'. The output is a table with three columns: VLAN Name, Status, and Ports.

VLAN Name	Status	Ports
1 default	active	Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/2
10 Computer_Eng_Students	active	
20 Computer_Eng_Staff	active	
30 Computer_Eng_Other_Devices	active	
40 EEE_Students	active	
50 EEE_Staff	active	
60 EEE_Other_Devices	active	
70 Civil_Eng_Students	active	
80 Civil_Eng_Staff	active	
90 Civil_Eng_Other_Devices	active	
100 Mech_Eng_Students	active	
110 Mech_Eng_Staff	active	
120 Mech_Eng_Other_Devices	active	
130 IDS_Students	active	
140 IDS_Staff	active	
150 IDS_Other_Devices	active	
160 Admin_Staff	active	
170 Admin_Printers	active	
180 CCTV_Cameras	active	Fa0/7
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

2. Computer Department L2 switch VLAN



The screenshot shows the CLI of a switch named 'Access switch computer'. The user has entered 'enable' and then 'show vlan brief'. The output is a table with three columns: VLAN Name, Status, and Ports.

VLAN Name	Status	Ports
1 default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/12, Fa0/13, Fa0/14, Fa0/18 Fa0/15, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/2
10 Com_Eng_Students	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5
20 Com_Eng_Staff	active	Fa0/10, Fa0/11
30 Com_Eng_Other_Devices	active	Fa0/15, Fa0/16, Fa0/17
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

3. EEE department L2 switch VLAN

Access switch EEE

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/15, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/15, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/16, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/16, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/17, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/17, changed state to up

Switch>enable
Switch#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/12, Fa0/13, Fa0/14, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/2
40 EEE_Students	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5
50 EEE_Staff	active	Fa0/10, Fa0/11
60 EEE_Other_Devices	active	Fa0/15, Fa0/16, Fa0/17
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

Switch#

Copy Paste

☐ Top

4. Civil department L2 switch VLAN

Access switch Civil

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/10, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/15, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/15, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/16, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/16, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/17, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/17, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/11, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/11, changed state to up

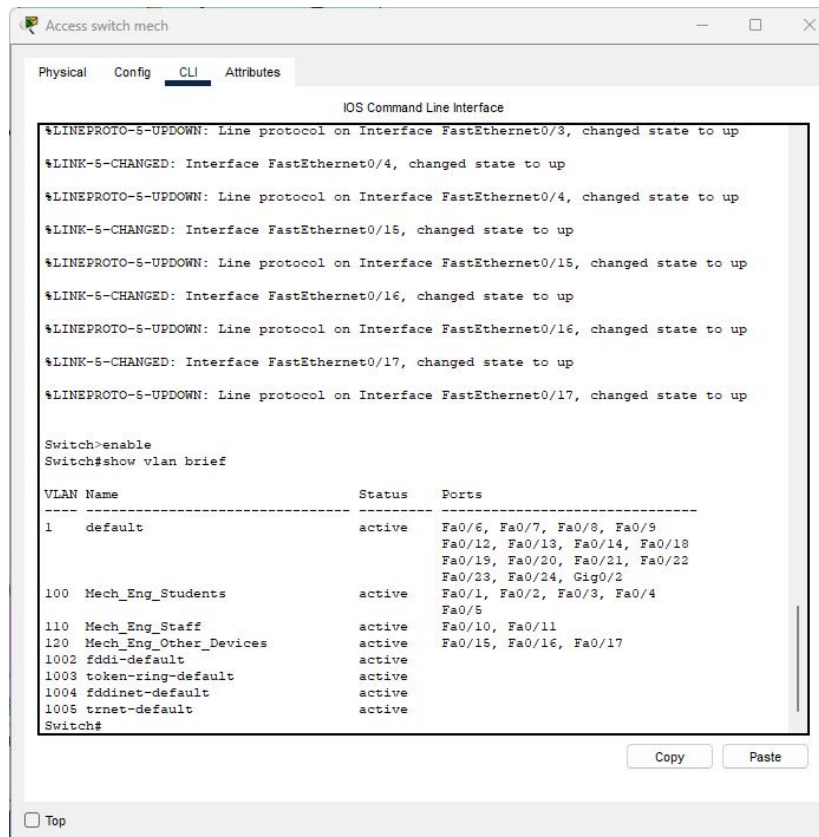
Switch>ENABLE
Switch#SHOW VLAN BRIEF
```

VLAN Name	Status	Ports
1 default	active	Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/2
70 Civil_Eng_Students	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5
80 Civil_Eng_Staff	active	Fa0/10, Fa0/11
90 Civil_Eng_Other_Devices	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

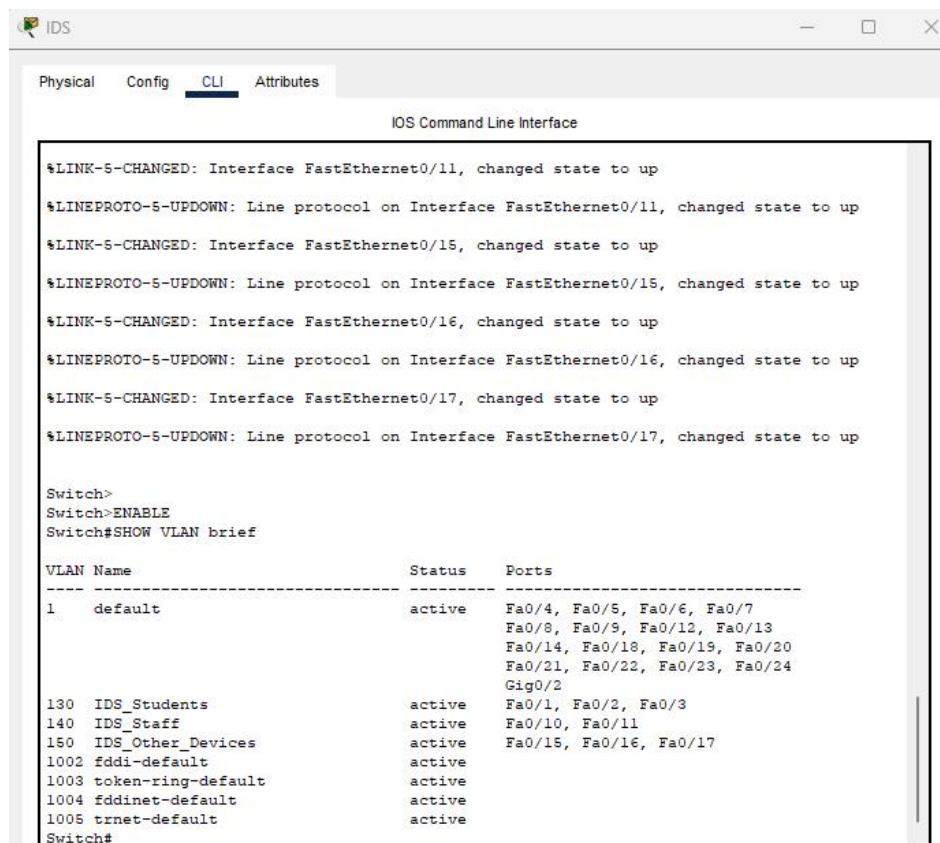
Switch#

Copy Paste

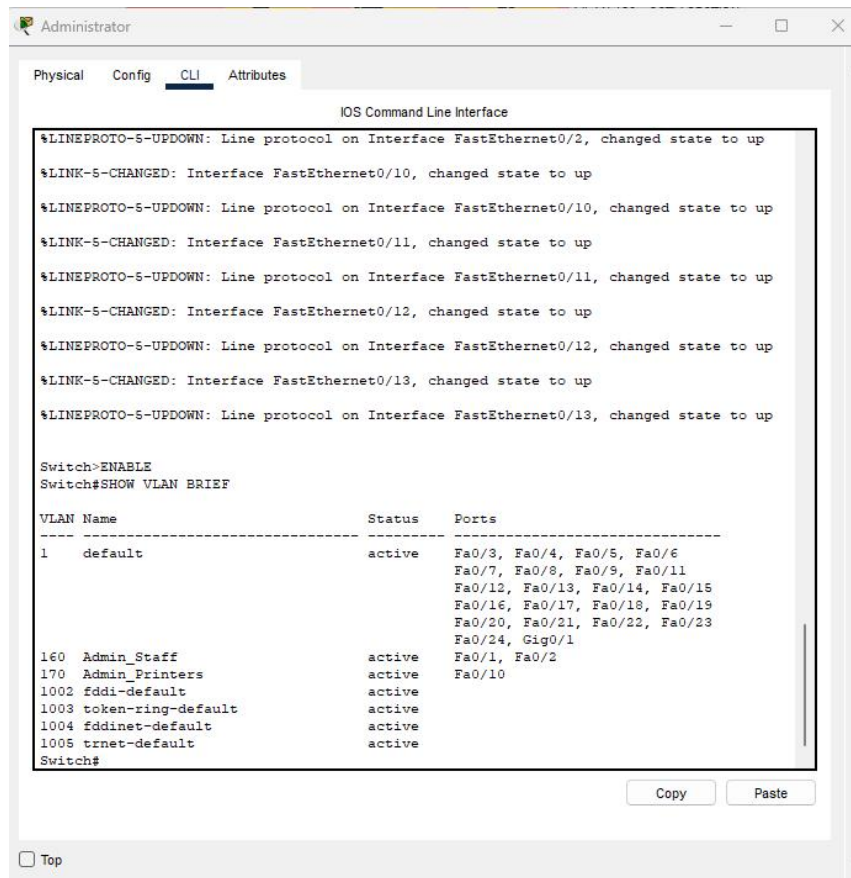
5. Mechanical department L2 switch VLAN



6. ADMINISTRATION department L2 switch VLAN



7. IDS L2 switch VLAN



The screenshot shows a network switch's CLI interface. The window title is "Administrator". The tabs are "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, showing the "IOS Command Line Interface".

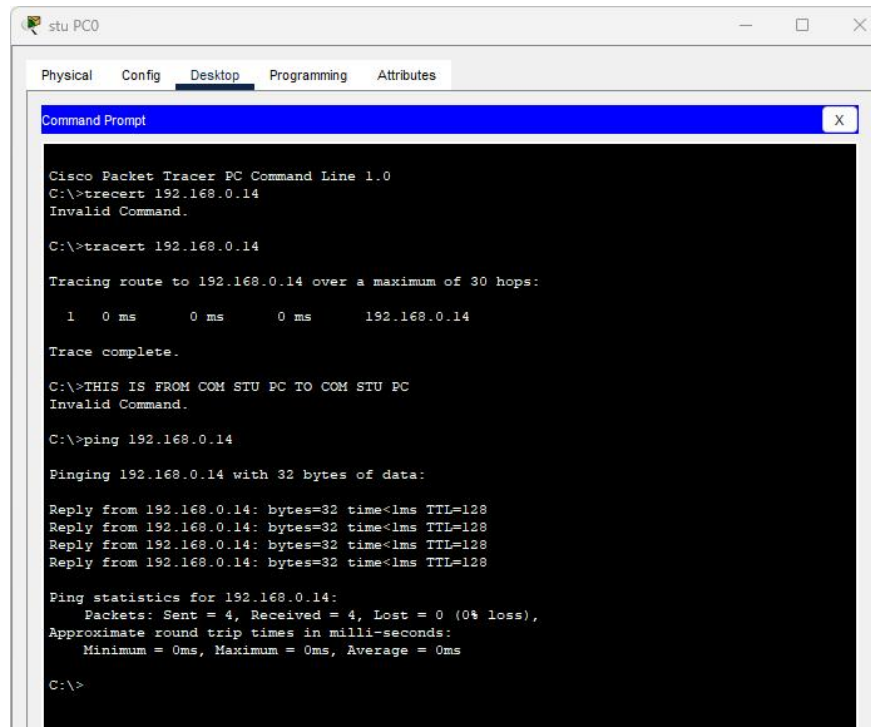
The CLI output shows the following commands and responses:

```
Switch>ENABLE
Switch#SHOW VLAN BRIEF
```

VLAN Name	Status	Ports
1 default	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1
160 Admin_Staff	active	Fa0/1, Fa0/2
170 Admin_Printers	active	Fa0/10
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

The CLI prompt is "Switch#". There are "Copy" and "Paste" buttons at the bottom right of the CLI window.

2.2 Device connectivity (ping and traceroute results)



```
stu PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>tracert 192.168.0.14
Invalid Command.

C:\>tracert 192.168.0.14

Tracing route to 192.168.0.14 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    192.168.0.14

Trace complete.

C:\>THIS IS FROM COM STU PC TO COM STU PC
Invalid Command.

C:\>ping 192.168.0.14

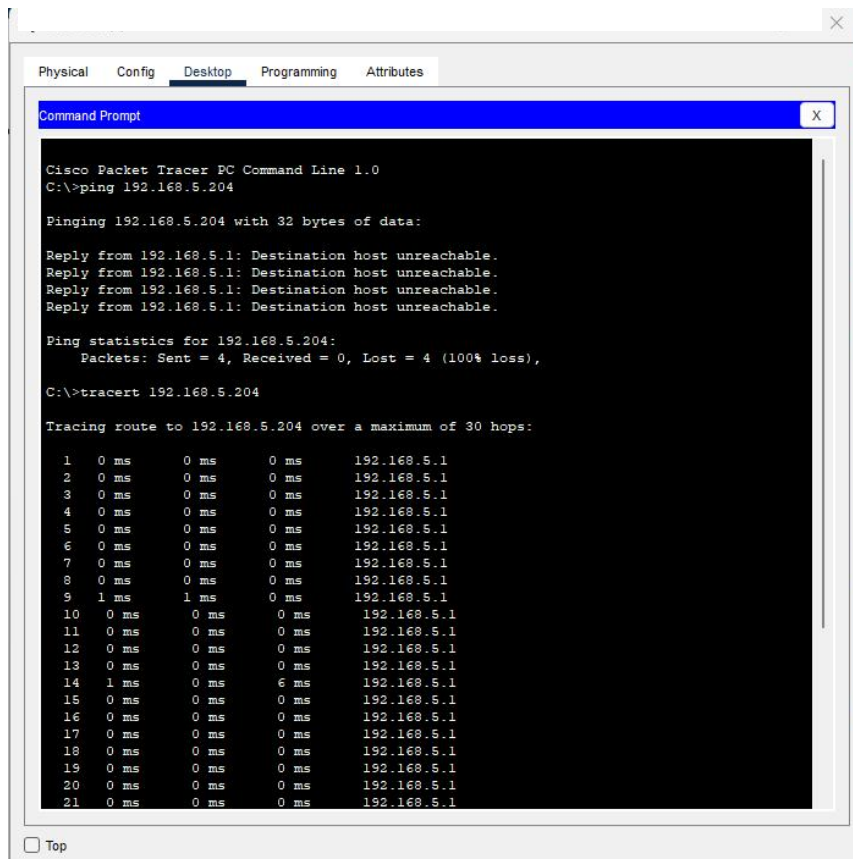
Pinging 192.168.0.14 with 32 bytes of data:

Reply from 192.168.0.14: bytes=32 time<1ms TTL=128
Reply from 192.168.0.14: bytes=32 time<1ms TTL=128
Reply from 192.168.0.14: bytes=32 time<1ms TTL=128
Reply from 192.168.0.14: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.0.14:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

From student pc to student pc : COMPUTER DEPARTMENT



```
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.5.204

Pinging 192.168.5.204 with 32 bytes of data:

Reply from 192.168.5.1: Destination host unreachable.
Reply from 192.168.5.1: Destination host unreachable.
Reply from 192.168.5.1: Destination host unreachable.
Reply from 192.168.5.1: Destination host unreachable.

Ping statistics for 192.168.5.204:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>tracert 192.168.5.204

Tracing route to 192.168.5.204 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    192.168.5.1
  2  0 ms    0 ms    0 ms    192.168.5.1
  3  0 ms    0 ms    0 ms    192.168.5.1
  4  0 ms    0 ms    0 ms    192.168.5.1
  5  0 ms    0 ms    0 ms    192.168.5.1
  6  0 ms    0 ms    0 ms    192.168.5.1
  7  0 ms    0 ms    0 ms    192.168.5.1
  8  0 ms    0 ms    0 ms    192.168.5.1
  9  1 ms    1 ms    0 ms    192.168.5.1
 10  0 ms    0 ms    0 ms    192.168.5.1
 11  0 ms    0 ms    0 ms    192.168.5.1
 12  0 ms    0 ms    0 ms    192.168.5.1
 13  0 ms    0 ms    0 ms    192.168.5.1
 14  1 ms    0 ms    6 ms    192.168.5.1
 15  0 ms    0 ms    0 ms    192.168.5.1
 16  0 ms    0 ms    0 ms    192.168.5.1
 17  0 ms    0 ms    0 ms    192.168.5.1
 18  0 ms    0 ms    0 ms    192.168.5.1
 19  0 ms    0 ms    0 ms    192.168.5.1
 20  0 ms    0 ms    0 ms    192.168.5.1
 21  0 ms    0 ms    0 ms    192.168.5.1

Top
```

From student pc to Printer : CIVIL DEPARTMENT

```
Physical  Config  Desktop  Programming  Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>PING 192.168.5.193

Pinging 192.168.5.193 with 32 bytes of data:

Reply from 192.168.5.193: bytes=32 time<1ms TTL=255
Reply from 192.168.5.193: bytes=32 time<1ms TTL=255
Reply from 192.168.5.193: bytes=32 time<1ms TTL=255
Reply from 192.168.5.193: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.5.193:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>TRACERT 192.168.5.193

Tracing route to 192.168.5.193 over a maximum of 30 hops:

  0  3 ms    0 ms    0 ms    192.168.5.193

Trace complete.

C:\>
```

From staff pc to Printer : CIVIL DEPARTMENT

```
STU PC15

Physical  Config  Desktop  Programming  Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.6.204

Pinging 192.168.6.204 with 32 bytes of data:

Reply from 192.168.6.1: Destination host unreachable.
Reply from 192.168.6.1: Destination host unreachable.
Reply from 192.168.6.1: Destination host unreachable.
Reply from 192.168.6.1: Destination host unreachable.

Ping statistics for 192.168.6.204:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

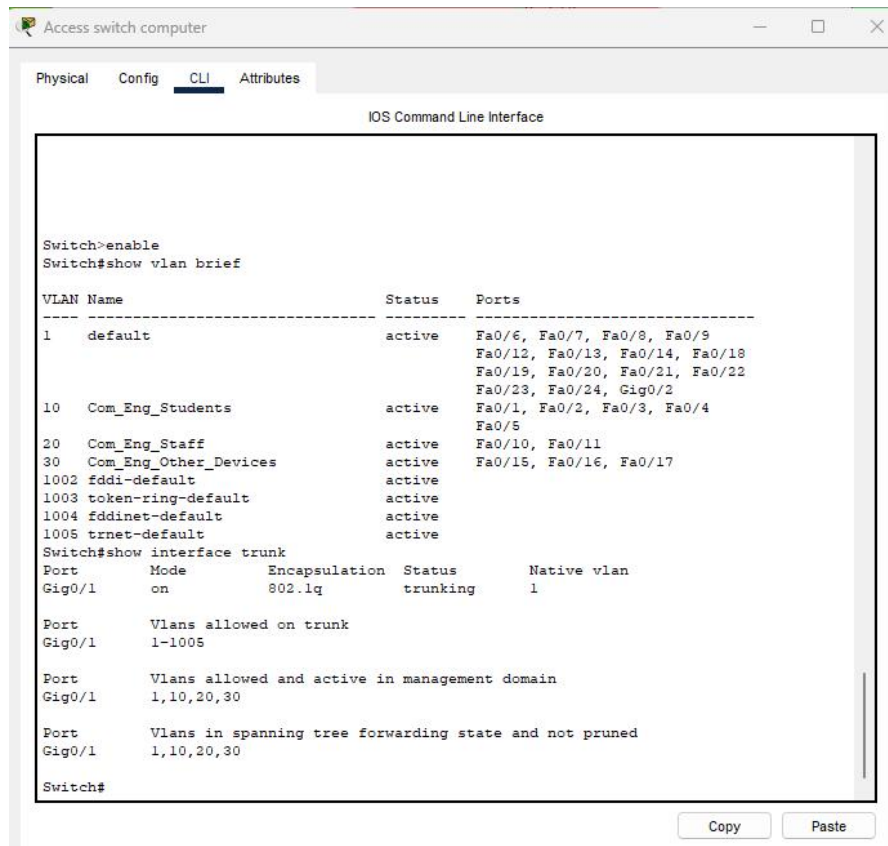
C:\>
C:\>tracert 192.168.6.204

Tracing route to 192.168.6.204 over a maximum of 30 hops:

  0  0 ms    0 ms    1 ms    192.168.6.1
  1  0 ms    0 ms    0 ms    192.168.6.1
  2  0 ms    0 ms    0 ms    192.168.6.1
  3  0 ms    0 ms    0 ms    192.168.6.1
  4  0 ms    0 ms    0 ms    192.168.6.1
  5  0 ms    0 ms    0 ms    192.168.6.1
  6  0 ms    0 ms    0 ms    192.168.6.1
  7  0 ms    0 ms    0 ms    192.168.6.1
  8  0 ms    17 ms   0 ms    192.168.6.1
  9  0 ms    0 ms    0 ms    192.168.6.1
 10  0 ms    0 ms    0 ms    192.168.6.1
 11  0 ms    0 ms    0 ms    192.168.6.1
 12  0 ms    0 ms    0 ms    192.168.6.1
 13  0 ms    0 ms    0 ms    192.168.6.1
 14  0 ms    0 ms    0 ms    192.168.6.1
 15  0 ms    0 ms    0 ms    192.168.6.1
 16  0 ms    0 ms    0 ms    192.168.6.1
 17  0 ms    0 ms    0 ms    192.168.6.1
 18  0 ms    0 ms    0 ms    192.168.6.1
 19  0 ms    0 ms    0 ms    192.168.6.1
 20  0 ms    0 ms    0 ms    192.168.6.1
```

From Student pc to Printer : MECHANICAL DEPARTMENT

2.3 VLAN and routing functionality tests



```
Switch>enable
Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                           Fa0/12, Fa0/13, Fa0/14, Fa0/18
                                           Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                           Fa0/23, Fa0/24, Gig0/2
10   Com_Eng_Students        active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                           Fa0/5
20   Com_Eng_Staff           active    Fa0/10, Fa0/11
30   Com_Eng_Other_Devices   active    Fa0/15, Fa0/16, Fa0/17
1002 fddi-default           active
1003 token-ring-default    active
1004 fddinet-default       active
1005 trnet-default         active

Switch#show interface trunk

Port      Mode      Encapsulation  Status      Native vlan
Gig0/1    on        802.1q         trunking    1

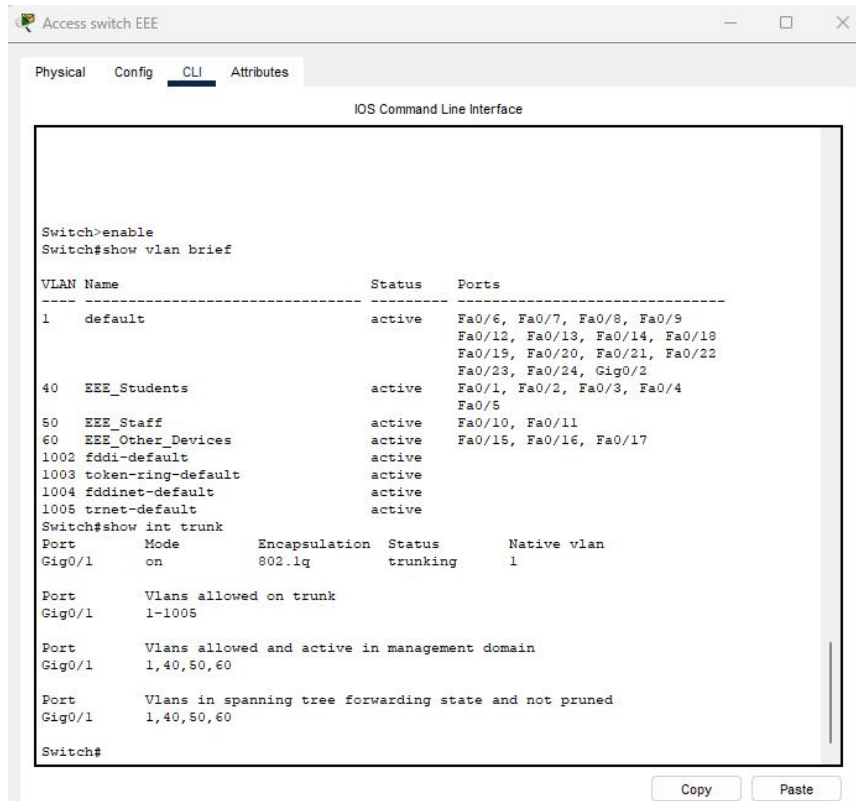
Port      Vlans allowed on trunk
Gig0/1    1-1005

Port      Vlans allowed and active in management domain
Gig0/1    1,10,20,30

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    1,10,20,30

Switch#
```

VLAN and routing functionality tests using
show vlan brief and show interfaces trunk commands. : COMPUTER DEPARTMENT



```
Switch>enable
Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                           Fa0/12, Fa0/13, Fa0/14, Fa0/18
                                           Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                           Fa0/23, Fa0/24, Gig0/2
40   EEE_Students            active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                           Fa0/5
50   EEE_Staff               active    Fa0/10, Fa0/11
60   EEE_Other_Devices       active    Fa0/15, Fa0/16, Fa0/17
1002 fddi-default           active
1003 token-ring-default    active
1004 fddinet-default       active
1005 trnet-default         active

Switch#show int trunk

Port      Mode      Encapsulation  Status      Native vlan
Gig0/1    on        802.1q         trunking    1

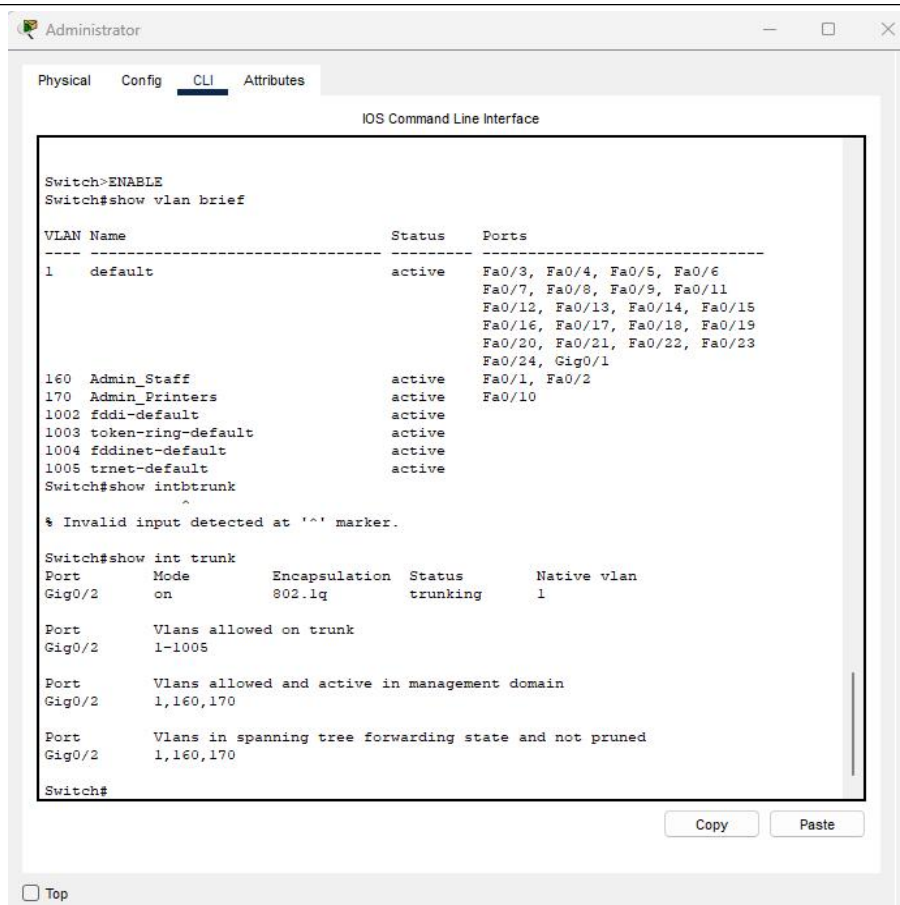
Port      Vlans allowed on trunk
Gig0/1    1-1005

Port      Vlans allowed and active in management domain
Gig0/1    1,40,50,60

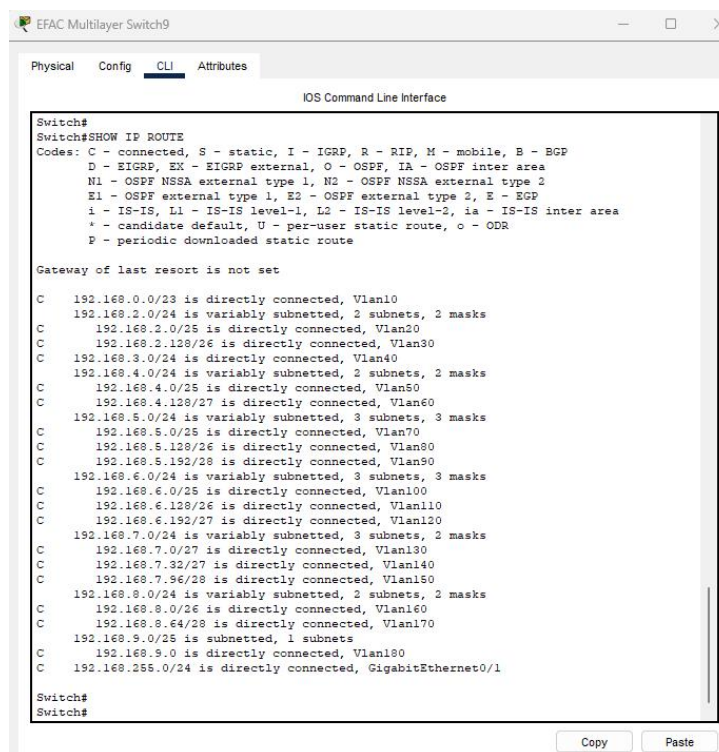
Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    1,40,50,60

Switch#
```

VLAN and routing functionality tests using
show vlan brief and show interfaces trunk commands. : EEE DEPARTMENT



VLAN and routing functionality tests using
show vlan brief and show interfaces trunk commands. : ADMINISTROTOR



VLAN and routing functionality tests using show IP route command

3. Scalability Validation

Sub net Mask & CIDR Notation

1. Example: **192.168.1.0/24**
2. CIDR /24 means **255.255.255.0** → 24 bits for the network, 8 bits for hosts

Number of Hosts in a Sub-nets

1. Formula: $2^n - 2$ (Subtract 2 for network and broadcast addresses)
2. Example: /24 has 8 host bits → $2^8 - 2 = 254$ hosts

Subnet Ranges & Broadcast Address

1. Example: **192.168.1.0/26** (Subnet Mask: 255.255.255.192)
2. Subnet Size: $2^6 = 64$ IPs
3. Usable Hosts: 62 (64 - 2 for network & broadcast)
4. First usable IP: 192.168.1.1
5. Last usable IP: 192.168.1.62
6. Broadcast IP: 192.168.1.63

Example: Computer Engineering - Students VLAN

- IP Address: 192.168.0.0/23
- Subnet Mask: 255.255.254.0
- CIDR Notation: /23
- Total Bits for Network: 23
- Total Bits for Hosts: $32 - 23 = 9$
- Total IPs in the Subnet: $2^9 = 512$
- Usable Hosts: $512 - 2 = 510$ (subtracting network and broadcast addresses)

Subnet Expansion for Future Growth

Each subnet has been planned with **30% additional capacity** to prevent immediate restructuring when new devices are added.

Example Calculation (Computer Engineering Students)

- Current Devices: 250
- Growth Factor (30%): $250 \times 1.3 = 325$

- **Subnet Selection:**
 - A **/24 subnet (256 hosts)** would be insufficient.
 - A **/23 subnet (512 hosts)** is selected to accommodate future expansion.

Device Growth Table

This table outlines the total devices, projected future growth (30%), and the required hosts for each department and device type.

Department	Device Type	Total Devices	Future Growth (30%)	Required Hosts	Department	Device Type	Total Devices
Computer Eng	Students	250	325	512	Computer Eng	Students	250
	Staff	50	65	128		Staff	50
	Other Devices	27	35	64		Other Devices	27
EE Eng	Students	150	195	256	EE Eng	Students	150
	Staff	50	65	128		Staff	50
	Other Devices	17	22	32		Other Devices	17
Civil Eng	Students	75	98	128	Civil Eng	Students	75
	Staff	25	33	64		Staff	25
	Other Devices	7	10	16		Other Devices	7
Mech Eng	Students	75	98	128	Mech Eng	Students	75
	Staff	25	33	64		Staff	25
	Other Devices	12	16	32		Other Devices	12
IDS	Students	15	20	32	IDS	Students	15
	Staff	25	33	64		Staff	25
	Other Devices	7	10	16		Other Devices	7

Summary of Added Devices and Network Performance

Added Devices

The network design includes the following devices:

- **Core Router:** Handles inter-VLAN routing and connectivity.
- **Core L3 Switch:** Connects department L3 switches and manages VLAN traffic.
- **Department L3 Switches:** One per department to handle VLANs and inter-VLAN routing.
- **L2 Switches:** Used for end-user connectivity (students, staff, printers, and other devices).
- **End-User Devices:** Student PCs, staff PCs, printers, and other networked equipment.
- **CCTV System:** A dedicated VLAN and switch for security cameras.

Network Performance

- **IP Addressing & Subnetting:** Proper allocation of subnets to ensure efficient IP management.
- **VLAN Implementation:** Optimized segmentation for security and performance.
- **Routing Efficiency:** Inter-VLAN routing enabled with a core router for seamless communication.
- **Traffic Management:** Access Control Lists (ACLs) prevent students from accessing unauthorized devices.
- **Scalability:** Designed with future growth in mind, accommodating a 30% expansion.
- **Reliability:** Redundant connections between core and department switches enhance network stability.

In My described network setup, where you have a main core router, a main core multilayer switch, and department-specific multilayer and Layer 2 (L2) switches, the performance would depend on various factors like bandwidth, network traffic, the scale of the network, and the configuration of the switches. Let's break it down based on the layers you've mentioned:

1. Core Router:

- This acts as the gateway to the external network (e.g., the internet) or inter-network communication between different departments or subnets.
- Performance-wise, the core router should be high-performance to handle routing between VLANs, and the data throughput should be high to avoid bottlenecks. If it's underpowered or misconfigured, it could cause delays or congestion.

2. Core Multilayer Switch:

- This switch combines both Layer 2 (switching) and Layer 3 (routing) functionalities.
- The core multilayer switch should have high throughput and support advanced features like VLAN routing, Quality of Service (QoS), and Access Control Lists (ACLs) to ensure smooth traffic flow between departments and prevent network congestion.
- Depending on the number of VLANs, routing protocols (like OSPF, EIGRP), and the amount of inter-department traffic, this switch will need to be adequately provisioned to avoid performance bottlenecks.

3. Department Multilayer Switches:

- Each department switch being a multilayer switch means it can handle both Layer 2 switching (within the department) and Layer 3 routing (between departments).
- These switches will handle local routing within the department as well as routing between departments (if any). Their performance will impact inter-department communication and internal data flow. Ensuring adequate CPU, RAM, and throughput capabilities on these switches is crucial for minimizing delays.
- The load on each switch will depend on the number of devices and the amount of traffic generated within the department.

4. Department L2 Switches:

- These switches will function purely in Layer 2 and are primarily used for internal communication within the department (switching traffic between devices in the same department).
- Performance here is less critical for inter-department traffic but should still be considered for managing VLANs and broadcast traffic within the department. Network segmentation using VLANs can help in reducing unnecessary traffic on L2 switches.

Potential Performance Considerations:

- **Traffic Load:** If the network traffic within or between departments is high, it might stress the core multilayer switch or the core router. You may need to implement QoS to prioritize critical traffic or use link aggregation to increase bandwidth.
- **Redundancy:** For high availability, redundancy with technologies like HSRP, VRRP, or GLBP should be considered for the core devices (router and multilayer switches). This ensures that a failure in one part doesn't bring down the entire network.
- **Latency:** If too many hops are involved (especially with multilayer switches doing routing), it could increase latency. Try to minimize unnecessary routing hops if possible.
- **Scalability:** As the network grows, ensuring that your multilayer switches can handle increased traffic without performance degradation is key. Ensuring adequate processing power (CPU and memory) and high-speed interfaces is essential.

4. Clarity and Consistency

ACL Configuration and Access Control

To enhance **network security** and ensure proper **network segmentation**, **Access Control Lists (ACLs)** were implemented across the network. These ACLs block access from the **Student VLANs** to the **Staff and Department VLANs**, preventing students from accessing sensitive resources in other departments. The rules are specifically applied on each departmental VLAN interface to filter inbound traffic from the student network.

ACL Explanation

Deny Rules:

1. Each departmental ACL restricts access from the **Student VLANs** (192.168.0.0/24, 192.168.3.0/24, etc.) to the **respective department VLANs** (e.g., 192.168.2.0/24, 192.168.4.0/24, etc.) using `deny` statements.
2. These **deny rules** ensure that students cannot access resources (such as servers, printers, or sensitive devices) in the staff or other department VLANs.

Permit Rule:

1. The `permit ip any any` rule allows all other traffic that is not specifically denied. This ensures that students can still access external resources, such as the internet, without being restricted by these ACLs.

Interface Application:

1. The `ip access-group BLOCK_STUDENTS in` command applies the ACL on each **department VLAN interface** to filter inbound traffic from the **Student VLAN**.
2. By applying the ACL to the VLAN interfaces (`Vlan10`, `Vlan40`, etc.), you're ensuring that these policies are enforced whenever traffic enters the VLAN from the student network.

Detailed Breakdown of Your Code

Computer Department:

1. Denies traffic from the student network (192.168.0.0/24) to specific subnets in the staff network (192.168.2.0/25 and 192.168.2.128/26).
2. Permits other traffic (like internet access).
3. ACL applied to **Vlan10**.

EEE Department:

1. Denies traffic from the student network (192.168.3.0/24) to specific subnets in the staff network (192.168.4.0/25 and 192.168.4.128/27).
2. Permits other traffic (like internet access).
3. ACL applied to **Vlan40**.

Civil Department:

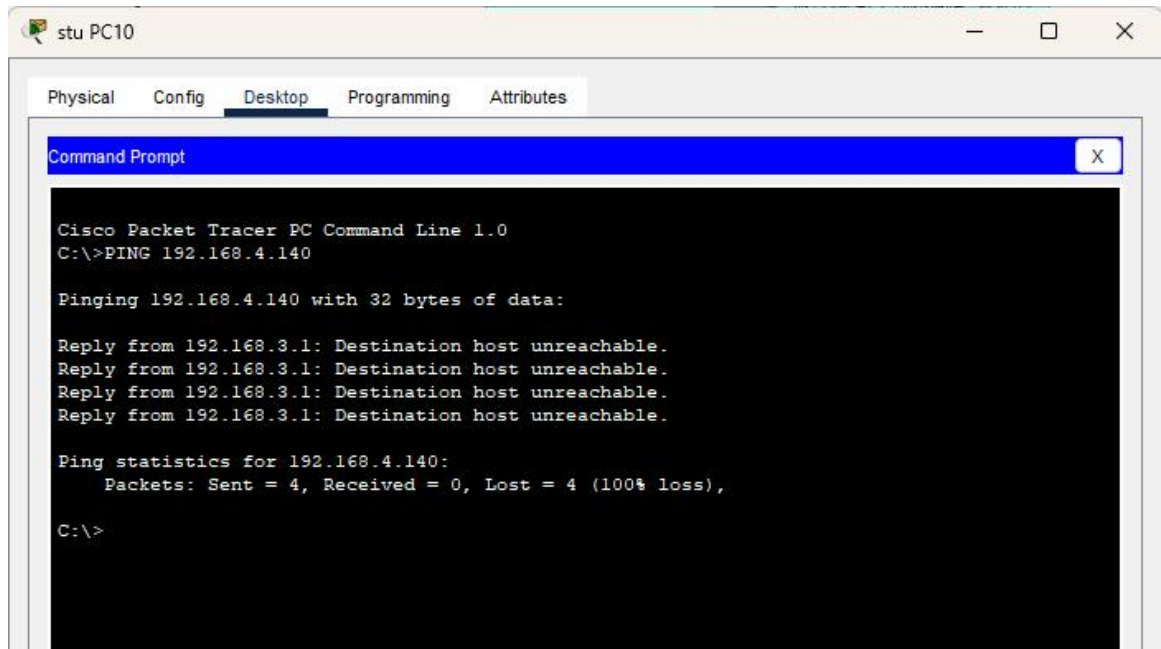
1. Denies traffic from the student network (192.168.5.0/25) to specific subnets in the staff network (192.168.5.128/26 and 192.168.5.192/28).
2. Permits other traffic (like internet access).
3. ACL applied to **Vlan70**.

Mechanical Department:

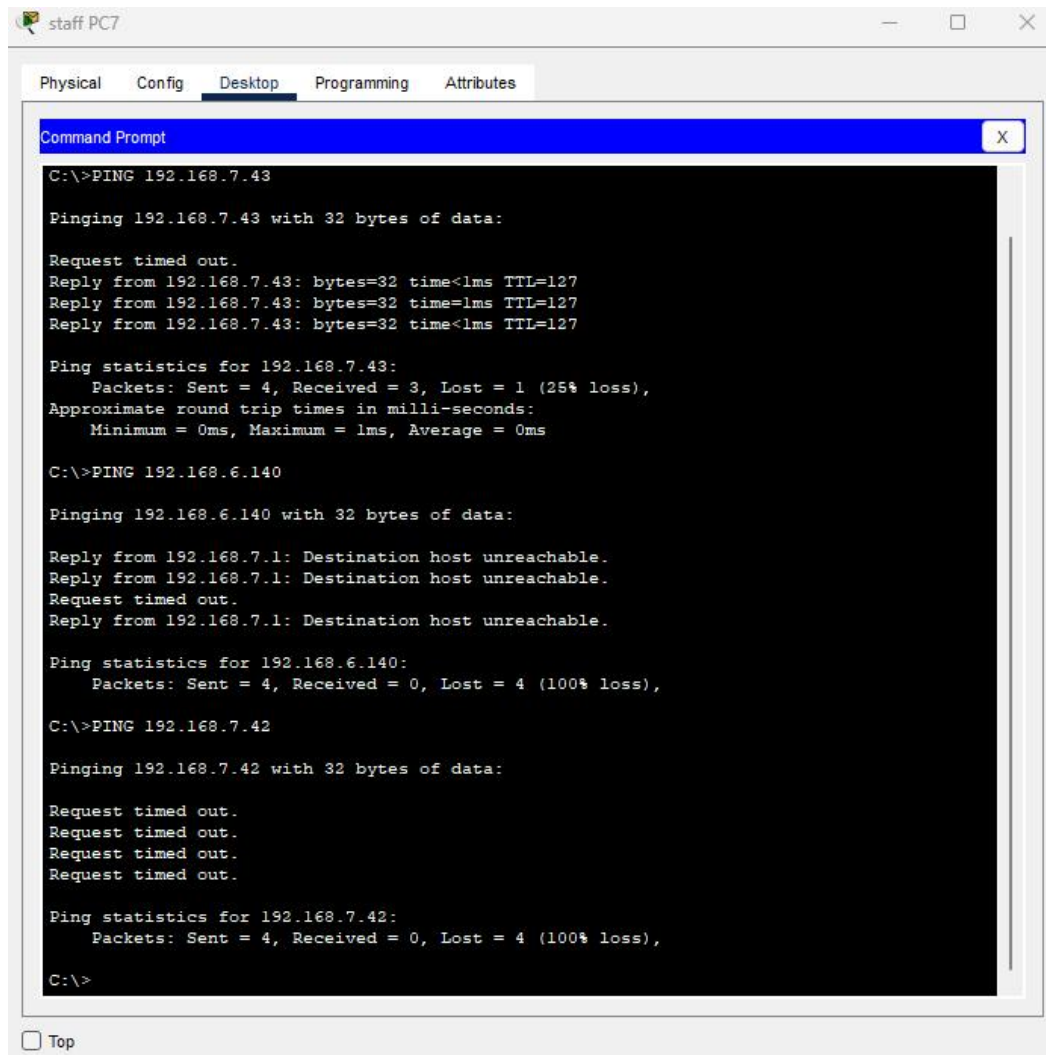
1. Denies traffic from the student network (192.168.6.0/25) to specific subnets in the staff network (192.168.6.128/26 and 192.168.6.192/27).
2. Permits other traffic (like internet access).
3. ACL applied to **Vlan100**.

IDS (Industrial Design and Simulation):

1. Denies traffic from the student network (192.168.7.0/27) to specific subnets in the staff network (192.168.7.32/27 and 192.168.7.96/28).
2. Permits other traffic (like internet access).
3. ACL applied to **Vlan130**.



Student Can't access the Printers or others devices. : EEE DEPARTMENT



Student Can't access the STAFF PCs. : MECHANICAL DEPARTMENT

Here are all the ACL configurations

Computer Department:

```
bash
CopyEdit
enable
configure terminal
ip access-list extended BLOCK_STUDENTS
deny ip 192.168.0.0 0.0.1.255 192.168.2.0 0.0.0.127
deny ip 192.168.0.0 0.0.1.255 192.168.2.128 0.0.0.63
permit ip any any exit
interface Vlan10
ip access-group BLOCK_STUDENTS inexit
write memory
```

EEE Department:

```
bash
CopyEdit
enable
configure terminal
ip access-list extended BLOCK_STUDENTS
deny ip 192.168.3.0 0.0.0.255 192.168.4.0 0.0.0.127
deny ip 192.168.3.0 0.0.0.255 192.168.4.128 0.0.0.31
permit ip any any exit
interface Vlan40
ip access-group BLOCK_STUDENTS inexit
write memory
```

Civil Department:

```
bash
CopyEdit
enable
configure terminal
ip access-list extended BLOCK_STUDENTS
deny ip 192.168.5.0 0.0.0.127 192.168.5.128 0.0.0.63
deny ip 192.168.5.0 0.0.0.127 192.168.5.192 0.0.0.15
permit ip any any exit
interface Vlan70
ip access-group BLOCK_STUDENTS inexit
write memory
```

Mechanical Department:

```
bash
CopyEdit
enable
configure terminal
ip access-list extended BLOCK_STUDENTS
deny ip 192.168.6.0 0.0.0.127 192.168.6.128 0.0.0.63
deny ip 192.168.6.0 0.0.0.127 192.168.6.192 0.0.0.31
```

```
permit ip any any exit
interface Vlan100
ip access-group BLOCK_STUDENTS inexit
write memory
```

IDS (Industrial Design and Simulation) Department:

```
bash
CopyEdit
enable
configure terminal
ip access-list extended BLOCK_STUDENTS
deny ip 192.168.7.0 0.0.0.31 192.168.7.32 0.0.0.63
deny ip 192.168.7.0 0.0.0.31 192.168.7.96 0.0.0.15
permit ip any any exit
interface Vlan130
ip access-group BLOCK_STUDENTS inexit
write memory
```

Check all codes I used here : https://github.com/Kosaladesh2002/Network_Project.git